Inorganic CUME

March 8, 2019

Note: Answer any five questions below. Only the first five of your answers will be graded.

1. (**20 points**) Basics for the powder XRD work on a powdery solid sample:
   1) Briefly discuss each parameter in the Bragg's Diffraction Equation (**5'**);
   2) Depending on the metal targets to generate X-rays, the resultant X-rays' wavelengths are $\lambda_{Cr} = 0.229$ (nm), $\lambda_{Fe} = 0.194$ (nm), $\lambda_{Co} = 0.179$ (nm), $\lambda_{Cu} = 0.154$ (nm), and $\lambda_{Mo} = 0.071$ (nm). Please explain
      a) why the λ reduces as the target atomic weight increases (**5'**)
      b) which X-ray is the most ionizing/harmful to human user (**5'**)
   3) Explain the working principle and popular applications of the small angle XRD (or SAXRD) (**5'**).

2. (**20 points**) What equations and what information from a polycrystal sample's powder XRD pattern can you use to identify the sample's d-space and average particle-size?

3. (**20 points**) Explain (or guess by showing your reasoning) the reasons to cause the powder XRD pattern's
   1) some peaks missing; (**10'**)
   2) all peaks broadening. (**10'**)

4. (**20 points**) One tool newly available on this campus is the X-ray micro-computed tomography (MicroCT) for nondestructively imaging a complex solid's inner 3D-microstructure (somewhat like the medical CT).
   1) Should the microCT use mainly the transmission mode and/or the diffraction mode? (**5'**)
   2) Show your reasons to choose what X-ray wavelength(s) and/or strategy for thus-imaging e.g.
      i. a sealed battery's inner microstructure in both mm- and μm-sizes; (**5'**)
      ii. a 3D-printed polymer device's inner surface microstructure in the μm-size; (**5'**)
      iii. a solid (cancer) tumor's 3D-microstructure in the single cell-size. (**5'**)

5. (**20 points**) The 2017 Nobel Prize in Chemistry was given to three scientists who developed cryo-electron microscopy (Cryo-EM) that uses an electron beam (its λ shorter than X-ray's) to solve structures of large proteins not easy to form large single crystals (as required in the single crystal XRD). Discuss (or predict)
   1) the similarities/differences between a cryo-EM and a regular high resolution TEM; (**5'**)
   2) advantage(s) of using the ultralow temperature (i.e. cryo) and electron beam (**5'**)
   3) disadvantages of using the cryo-EM. (**10'**)

6. (**20 points**) Explain how to use the point group theory* to help you predict the general difference in ***IR spectrum*** between a tetrahedral complex cation and an octahedral complex anion.

7. (**20 points**) Explain how to use the point group theory* to help you predict the general difference in ***Raman spectrum*** between a tetrahedral complex cation and an octahedral complex anion.

8. (**20 points**) Explain how to use the point group theory* to help you predict the general difference in ***NMR spectrum*** between a tetrahedral complex cation and an octahedral complex anion.

9. (**20 points**) Explain how to use the point group theory* to help you predict the general difference in ***EPR spectrum*** between a tetrahedral complex cation and an octahedral complex anion.

10. (**20 points**) Explain how to use the point group theory* to help you predict the general difference in ***electronic absorption spectrum*** between a tetrahedral complex cation and an octahedral complex anion.

*The character tables are attached on the next page.

The character tables for the Inorganic CUME.

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$		
A_1	1	1	1	1	1		$x^2 + y^2 + z^2$
A_2	1	1	1	-1	-1		
E	2	-1	2	0	0		$(2z^2 - x^2 - y^2, x^2 - y^2)$
T_1	3	0	-1	1	-1	(R_x, R_y, R_z)	
T_2	3	0	-1	-1	1	(x, y, z)	(xy, xz, yz)

O_h	E	$8C_3$	$6C_2$	$6C_4$	$3C_2(= C_4^2)$	i	$6S_4$	$8S_6$	$3\sigma_h$	$6\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		x2 + y2 + z2
A_{2g}	1	1	-1	-1	1	1	-1	1	1	-1		
E_g	2	-1	0	0	2	2	0	-1	2	0		(2z2 - x2y2, x2 - y2)
T_{1g}	3	0	-1	1	-1	3	1	0	-1	-1	(Rx, Ry, Rz)	
T_{2g}	3	0	1	-1	-1	3	-1	0	-1	1		(xz, yz, xy)
A_{1u}	1	1	1	1	1	-1	-1	-1	-1	-1		
A_{2u}	1	1	-1	-1	1	-1	1	-1	-1	1		
E_u	2	-1	0	0	2	-2	0	1	-2	0		
T_{1u}	3	0	-1	1	-1	-3	-1	0	1	1	(x, y, z)	
T_{2u}	3	0	1	-1	-1	-3	1	0	1	-1		